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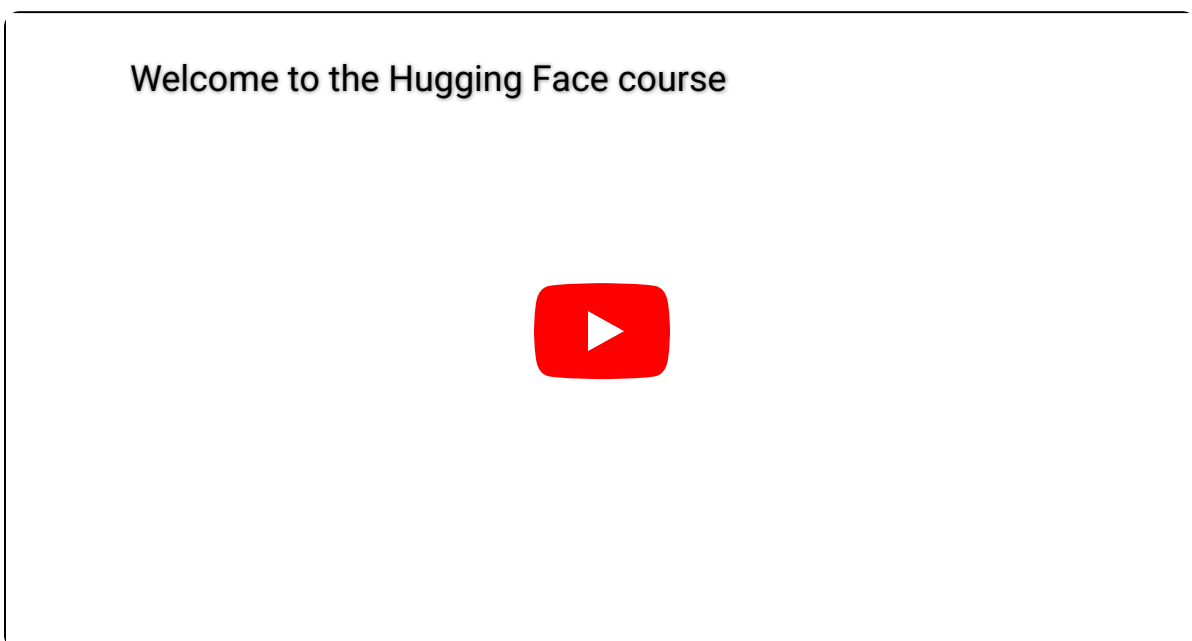
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Introduction

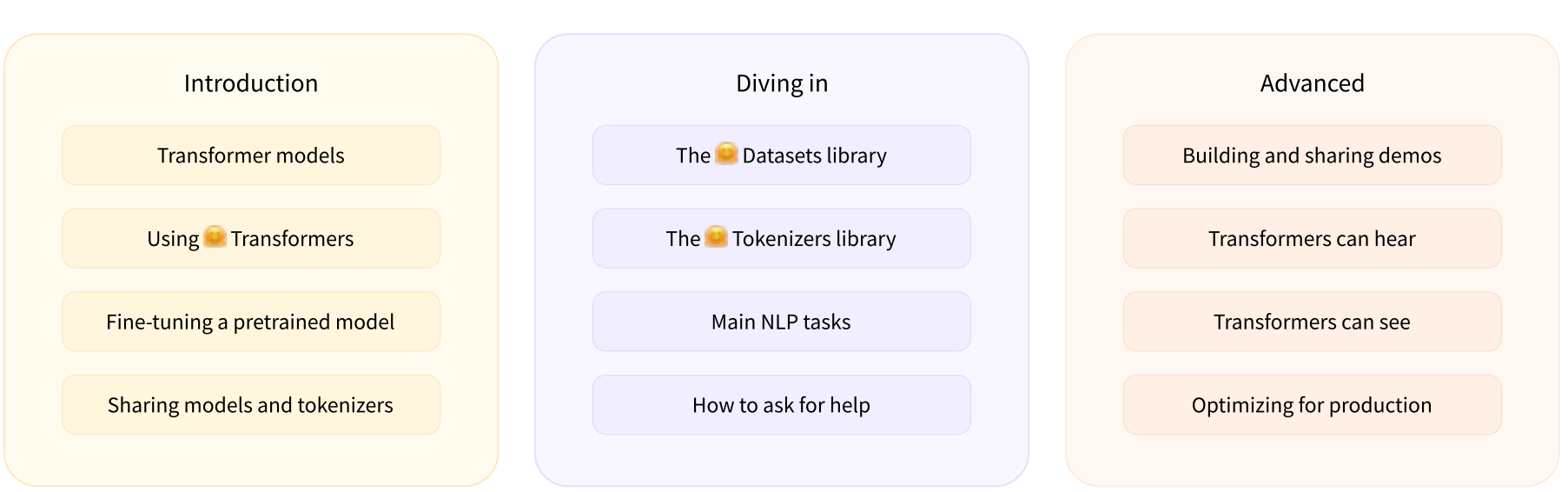
Welcome to the Course!



This course will teach you about natural language processing (NLP) using libraries from the Hugging Face ecosystem — Transformers, Datasets, Tokenizers, and Accelerate — as well as the Hugging Face Hub. It's completely free and without ads.

What to expect?

Here is a brief overview of the course:



- Chapters 1 to 4 provide an introduction to the main concepts of the Transformers library. By the end of this part of the course, you will be familiar with how Transformer models work and will know how to use a model from the Hugging Face Hub, fine-tune it on a dataset, and share your results on the Hub!
- Chapters 5 to 8 teach the basics of Datasets and Tokenizers before diving into classic NLP tasks. By the end of this part, you will be able to tackle the most common NLP problems by yourself.
- Chapters 9 to 12 go beyond NLP, and explore how Transformer models can be used to tackle tasks in speech processing and computer vision. Along the way, you'll learn how to build and share demos of your models, and optimize them for production environments. By the end of this part, you will be ready to apply Transformers to (almost) any machine learning problem!

This course:

- Requires a good knowledge of Python
- Is better taken after an introductory deep learning course, such as fast.ai's Practical Deep Learning for Coders or one of the programs developed by DeepLearning.AI
- Does not expect prior PyTorch or TensorFlow knowledge, though some familiarity with either of those will help

After you've completed this course, we recommend checking out DeepLearning.AI's Natural Language Processing Specialization, which covers a wide range of traditional NLP models like naïve Bayes and LSTMs that are well worth knowing about!

Who are we?

About the authors:

Abubakar Abid completed his PhD at Stanford in applied machine learning. During his PhD, he founded Gradio, an open-source Python library that has been used to build over 600,000 machine learning demos. Gradio was acquired by Hugging Face, which is where Abubakar now serves as a machine learning team lead.

Matthew Carrigan is a Machine Learning Engineer at Hugging Face. He lives in Dublin, Ireland and previously worked as an ML engineer at Parse.ly and before that as a post-doctoral researcher at Trinity College Dublin. He does not believe we're going to get to AGI by scaling existing architectures, but has high hopes for robot immortality regardless.

Lysandre Debut is a Machine Learning Engineer at Hugging Face and has been working on the Transformers library since the very early development stages. His aim is to make NLP accessible for everyone by developing tools with a very simple API.

Sylvain Gugger is a Research Engineer at Hugging Face and one of the core maintainers of the Transformers library. Previously he was a Research Scientist at fast.ai, and he co-wrote Deep Learning for Coders with fastai and PyTorch with Jeremy Howard. The main focus of his research is on making deep learning more accessible, by designing and improving techniques that allow models to train fast on limited resources.

Dawood Khan is a Machine Learning Engineer at Hugging Face. He's from NYC and graduated from New York University studying Computer Science. After working as an iOS Engineer for a few years, Dawood quit to start Gradio with his fellow co-founders. Gradio was eventually acquired by Hugging Face.

Merve Noyan is a developer advocate at Hugging Face, working on developing tools and building content around them to democratize machine learning for everyone.

Lucile Saulnier is a machine learning engineer at Hugging Face, developing and supporting the use of open source tools. She is also actively involved in many research projects in the field of Natural Language Processing such as collaborative training and BigScience.

Lewis Tunstall is a machine learning engineer at Hugging Face, focused on developing open-source tools and making them accessible to the wider community. He is also a co-author of the O'Reilly book Natural Language Processing with Transformers.

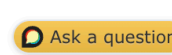
Leandro von Werra is a machine learning engineer in the open-source team at Hugging Face and also a co-author of the O'Reilly book Natural Language Processing with Transformers. He has several years of industry experience bringing NLP projects to production by working across the whole machine learning stack..

FAQ

Here are some answers to frequently asked questions:

- Does taking this course lead to a certification?** Currently we do not have any certification for this course. However, we are working on a certification program for the Hugging Face ecosystem — stay tuned!
- How much time should I spend on this course?** Each chapter in this course is designed to be completed in 1 week, with approximately 6-8 hours of work per week. However, you can take as much time as you need to complete the course.
- Where can I ask a question if I have one?** If you have a question about any section of the course, just click on the "Ask a question" banner at the top of the page to be automatically redirected to the right section of the Hugging Face forums:

Natural Language Processing

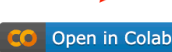
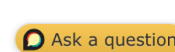


Before jumping into Transformer models, let's do a quick overview of what natural language processing is and why we care about it.

Note that a list of project ideas is also available on the forums if you wish to practice more once you have completed the course.

- Where can I get the code for the course?** For each section, click on the banner at the top of the page to run the code in either Google Colab or Amazon SageMaker Studio Lab:

Transformers, what can they do?



In this section, we will look at what Transformer models can do and use our first tool from the Transformers library: the pipeline() function.

The Jupyter notebooks containing all the code from the course are hosted on the huggingface/notebooks repo. If you wish to generate them locally, check out the instructions in the course repo on GitHub.

- How can I contribute to the course?** There are many ways to contribute to the course! If you find a typo or a bug, please open an issue on the course repo. If you would like to help translate the course into your native language, check out the instructions here.
- What were the choices made for each translation?** Each translation has a glossary and TRANSLATING.txt file that details the choices that were made for machine learning jargon etc. You can find an example for German here.
- Can I reuse this course?** Of course! The course is released under the permissive Apache 2 license. This means that you must give appropriate credit, provide a link to the license, and indicate if changes were made. You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use. If you would like to cite the course, please use the following BibTeX:

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Let's Go

Are you ready to roll? In this chapter, you will learn:

- How to use the pipeline() function to solve NLP tasks such as text generation and classification
- About the Transformer architecture
- How to distinguish between encoder, decoder, and encoder-decoder architectures and use cases

